Real Time Neuron Labeling

A practical architecture for streaming identity tracking, label vectors, and active probing

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Included: runnable synthetic demo (Python) packaged with this paper

Abstract

Real time neuron labeling is the problem of continuously assigning a stable identity and a functional description to each observed neuron as data streams in. In biology, full brain coverage is limited by access, physics, and safety, so a realistic target is dense but partial coverage with truthful labels expressed as a vector of properties plus uncertainty rather than a single name. This paper packages a minimal end to end architecture: online unitization, identity tracking, streaming label inference, and closed loop probing that actively selects the next stimulus to reduce uncertainty. A small open demo is provided that simulates hundreds of neurons with hidden tuning and produces live label snapshots.

1. Problem statement

The phrase label each neuron often implies a single human word that stays correct across all contexts. For most biological neurons, this is not a faithful representation. A neuron can participate in multiple computations, and its apparent role can change with state, task, and learning. Therefore, the output of a labeling system should be a structured profile: stimulus and state tuning, coupling to behavioral variables, and causal effect estimates when perturbations are available, each with confidence.

2. Architecture

Real time labeling decomposes into four streaming blocks. First, acquisition produces raw signals from electrodes or optical imaging. Second, online unitization converts signals into per unit activity traces, such as spike times or deconvolved events. Third, identity tracking maintains stable neuron ids across time despite drift in waveforms, motion, or extraction masks. Fourth, online labeling maps each stable id to a label vector and uncertainty, and optionally drives an active prober that selects the next stimulus or perturbation to reduce uncertainty.

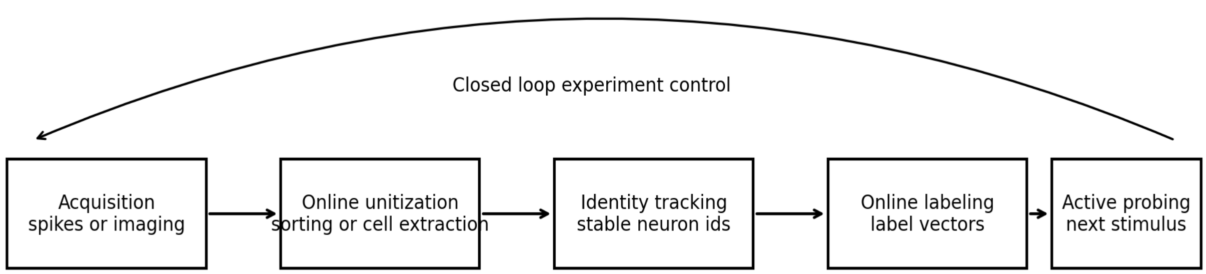

Figure 1. Streaming pipeline with a feedback loop from probing to acquisition.

3. What makes it real time

Real time in this context means three things. First, each processing stage runs incrementally on the newest data without waiting for the full session. Second, latency is low enough to support closed loop experiments, where stimulation or stimulus choice depends on recent activity. Third, identity is preserved online so that neuron level statistics accumulate rather than reset. Recent work demonstrates practical online pipelines for calcium imaging analysis (OnACID, CaImAn) and modern spike sorting frameworks (Kilosort, Kilosort4), while new real time spike sorting algorithms explicitly target closed loop latency budgets.

4. Output format: NeuronCards

A neuron label should be machine actionable and human readable. We recommend a NeuronCard schema per stable id that includes rate statistics, stimulus and state preference distributions, coupling to behavioral variables, and if available causal effects from stimulation or silencing. A single text label can be derived from the NeuronCard, but only as a lossy summary.

Example real time label snapshot

id	rate_hz	stim_pref	stim_conf	state_pref	state_conf	move_score
37	3.5	0	0.53	0	0.65	0.0
65	4.8	7	0.48	3	0.56	0.0
19	7.9	4	0.44	0	0.62	0.0
94	4.9	6	0.44	2	0.64	0.0

Figure 2. Example label snapshot table from the included synthetic demo.

5. Identity tracking

Identity tracking links an observed unit at time t to the same biological or simulated neuron at time t+1. In electrodes, waveforms drift and overlap, and in imaging, the field of view can shift or masks can deform. A minimal practical approach is feature based matching with an assignment solver, which is also the strategy used in the included demo. In real systems, identity tracking is often integrated into the unitization stage itself.

6. Active probing

Passive observation converges slowly when many stimuli and internal states exist. Active probing makes the problem tractable by selecting the next stimulus, context, or perturbation that maximally reduces uncertainty in the label vector. Closed loop optogenetic control and closed loop all optical manipulation show that activity guided stimulation is feasible, and provide the conceptual template for uncertainty driven probing.

7. Included demo

The demo implements the pipeline on synthetic data. A simulator generates spike counts for a population with hidden tuning to discrete stimuli and internal states, plus drifting per unit feature vectors. An identity tracker matches observed units to stable ids each step. An online labeler estimates stimulus and state preference distributions and reports confidences. An active prober selects the next stimulus that increases expected information gain.

Run commands (Windows PowerShell):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
python src\run_console.py
streamlit run src\run_dashboard.py
```

8. Limits and honest scaling

This paper does not claim that we can label every neuron in a human brain in real time with current technology. The point is to clarify the correct output type and the correct control loop. Coverage is bounded by physical access and safety. Even with perfect recording, causal labeling does not scale by testing neurons one by one; it scales by building models that generalize from cell type, connectivity, and partial interventions, and by reporting uncertainty.

References

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